

Project Proposal: Video Emotion Recognition in DUUI

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1 Issue of the Task

Emotion recognition in video streams has traditionally relied on isolated, unimodal analyses (e.g., analyzing only facial frames). However, human emotion—especially within complex, dynamic environments like political speeches or parliamentary debates—is inherently multimodal. The core issue this project addresses is the implementation and evaluation of a robust, multimodal emotion recognition component within the DUUI framework.

Initial testing with classic Vision Transformers (e.g., ViT-Base models trained on FER-2013 and AffectNet) revealed severe contextual biases. Specifically, head posture in video streams artificially skewed predictions, resulting in a $\sim 75\%$ “sad” classification regardless of the underlying training data distribution. This highlights the inadequacy of purely frame-based facial analysis. To resolve this, the project will investigate the intersection of video (non-verbal cues, facial action units, pose), audio (intonation), and text (semantic sentiment), evaluating whether modern multimodal LLMs (e.g., Phi-4, Qwen) can better resolve cross-modal conflicts and temporal inconsistencies than classic architectures.

2 Available Resources

The following resources and technology stacks are available and will be utilized:

- **Framework:** DUUI.
- **Classic ViT Models:** `trpakov/vit-face-expression` ($\sim 86\text{M}$ params, FER-2013, 7 classes) and `dima806/facial_emotions_image_detection` (AffectNet, 8 classes).
- **Multimodal LLMs:** Phi-4, Qwen.
- **Metrics & Taxonomies:** Ekman’s basic emotions (Anger, Disgust, Fear, Happiness, Sadness, Surprise, Contempt), Valence/Arousal continuous scales (-1 to 1), and FACS (Facial Action Coding System, ~ 40 Action Units).
- **Tech Stack:** Python, HuggingFace Transformers, REST APIs.

3 Requirements

To successfully deploy this component, the following requirements must be met:

- **Multimodal Integration:** The system must ingest and synchronize video, audio, and text streams within DUUI.
- **Architectural Comparison:** A formal evaluation comparing classic ViT models against multimodal LLMs in terms of temporal consistency and accuracy.

- **Multi-Level Analysis:** The system must categorize emotions across three distinct temporal levels:
 - *Micro:* Emotion per frame/segment (e.g., AU-based micromimicry).
 - *Meso:* Speech dynamics per speaker (e.g., duration of emotional states, rapid shifts).
 - *Macro:* Session context or daily progression (e.g., emotional escalation or contagion across multiple speakers).
- **Error Analysis:** The implementation must track divergence between sentiment and emotion (e.g., neutral text but emotionally charged audio).

4 Use Cases

1. **Cross-Modal Conflict Detection in Political Contexts:** Detecting instances where modalities contradict, such as a politician displaying “controlled anger” (neutral text, aggressive AUs, tense intonation). The system will determine if an LLM can resolve this conflict better than a strict rule-based model.
2. **Speaker Profiling & Stance Detection:** Analyzing a speaker’s baseline “emotional style” to determine their stance (supportive, opposing, neutral) based on the consistency of high valence/arousal and body language (e.g., open posture correlating with positive valence).
3. **Temporal Speech Dynamics & Escalation Tracking:** Monitoring a parliamentary session to detect emotional contagion. The system will track if a highly charged speech influences the emotional baseline of subsequent speakers.

5 Implementation Plan

The implementation is divided into four main phases:

- **Phase 1: Baseline Setup & Integration.** Integrate the two ViT-Base models into DUUI. Establish the data pipeline for frame extraction and map outputs to Ekman classes, Valence/Arousal, and AUs.
- **Phase 2: Multimodal Pipeline.** Integrate audio processing and NLP (text sentiment) via REST. Implement the Micro, Meso, and Macro temporal tracking structures.
- **Phase 3: LLM Integration.** Connect Phi-4/Qwen to act as the multimodal reasoning engine, feeding it the combined modality inputs to resolve conflicts.
- **Phase 4: Evaluation & Tuning.** Conduct error analysis specifically focusing on the previously identified head-pose bias. Benchmark the LLM’s conflict resolution against the classic models.

6 Project Schedule

The following schedule outlines the progression to the final presentation in mid-August 2026.

